

- 11 Mars has a diameter that is approximately 0.5 times the diameter of the Earth, and the mass of Mars is 0.1 times the mass of the Earth.

Given that the gravitational potential at the Earth's surface is -63 MJ kg^{-1} .

What is an approximate value for the gravitational potential at the surface of Mars?

- A** -13 MJ kg^{-1} **B** -25 MJ kg^{-1} **C** -95 MJ kg^{-1} **D** -320 MJ kg^{-1}

Ans: A

The gravitational potential at the surface of a planet is directly proportional to the planet's mass, and inversely proportional to its radius (or diameter).

$$\phi = -\frac{GM}{r}$$

$$\phi_{\text{earth}} = -\frac{GM_{\text{earth}}}{r_{\text{earth}}}$$

$$\phi_{\text{Mars}} = -\frac{G(0.1M_{\text{earth}})}{(0.5r_{\text{earth}})} = \frac{1}{5}(-63) = -12.6 \text{ MJ kg}^{-1}$$

Hence, the potential at Mars' surface is approximately -13 MJ kg^{-1} .

12 Container X contains neon and container Y contains argon. The two containers are identical